

Assignment 2

Topic: *Integrating and analysing multi-modal data and applying advanced techniques such as generative models and transfer learning.*

Programming Questions

Instruction: In this assignment, you can choose whether to use tensorflow or pytorch library. Your codes must be executable. If your code isn't developed in the google colab environments, you need to include the requirement file.

Questions

1) This task is about the zero-shot inference using CLIP. For this task, you are expected to use the CLIP API with the ViT-b 16 model. Use the MNIST dataset which can be downloaded directly from the torchvision library and present the 10-digit classification problem. This task doesn't involve any training process and you can use the CLIP model directly for zero-shot inferences where the text token might include the class names.

[30 marks]

2) This task is about the multi-modal model, CLIP which combines the text and image data. Develop the CLIP model consisting of the image encoder and the text encoder. Both encoders are pretrained. Train them using the contrastive loss in the downstream task, the flickr30k dataset (<https://huggingface.co/datasets/nlphuji/flickr30k>). To do so, you need to have adapters dynamically adjusted while leaving the encoder parameters fixed. Once training the CLIP model into the downstream task, you need to evaluate the model into the testing dataset. That is, your dataset should be partitioned into the training set and the evaluation set. The inference can be done by checking the class maximizing the similarity between the text features and the visual features.

[70 marks]

3) This task is about the transfer learning problem. Import the pretrained vgg16 model and perform the transfer learning problem to the fashion MNIST dataset (<https://docs.pytorch.org/vision/0.19/generated/torchvision.datasets.FashionMNIST.html>). Note that this model has been pretrained in the imagenet problem, thus having decent representations. To perform the transfer learning task, the fully connected layer must be replaced with new ones. The training process is done only to the new layers while fixing the other parameters. The number of output node should match to the number of classes of the fashion MNIST dataset.

[20 marks]

4) This task is still about the transfer learning problem in the fashion mnist dataset. Import the vgg16 model. This task differs from the previous task where you need to import the untrained vgg16 model. It is your choice whether you directly import the model from the library or craft your own vgg16 model. Note that the fully connected layer must be adapted to the fashion mnist problem having 10 classes and 28*28 image. Once creating the model, train the model from scratch encompassing all network parameters.

[20 marks]

5) Comment on your finding to the previous two tasks. Which approach gives you better result, i.e., training from scratch or transfer learning. Discuss the rationale of your finding. Your comments should be written in the same file as your codes.

[10 marks]

6) Design a generative adversarial network (GAN) to mimic the fashion mnist dataset. Note that GAN consists of generator and discriminator which can be customized depending on the problem at hand. Both networks are trained simultaneously where the generator is trained to fool the discriminator while the discriminator tries to distinguish whether an image is real or fake. To measure the quality of the generated images, don't forget to save the generated images into a folder. Hyper-parameters are left to your selection.

[50 marks]

Theoretical Questions

7) Discuss the masked self-attention layer in GPT2. Why is such mechanism powerful to address the NLP problem.

[20 marks]

8) Discuss the pre-training process in BERT. Why does the pretraining process improve BERT's performances to perform downstream tasks?

[20 marks]

9) Explain how you do the fine-tuning mechanism in CLIP? Essentially, the CLIP model is only for the zero-shot prediction. The transfer learning approach is done to apply CLIP to downstream tasks.

[20 marks]

10) Discuss the reason why the pretrained model usually performs better than the model trained from scratch. What does dataset the community in computer vision use to pretrain their model and explain the pretraining step for a vision model.

[20 marks]

11) Explain the generative adversarial network (GAN) including its training process and network structure. Your answer should include the loss functions of the generator and the discriminator.

[20 marks]

Submission Instruction: your codes and answers to theoretical questions need to be submitted online. If they comprise more than one file, you can put it into a single folder and zip it.